

Multimodal Translation and Mapping

Unit 3



Multimodal translation and mapping

Multimodal Translation & Mapping refers to converting or aligning information **across different modalities** — such as **text, images, audio, and video** — so that models can **understand, relate, and translate** between them.

Why it matters

- Language alone often loses meaning (e.g., describing visual scenes).
- Mapping across modalities enables models to:
 - ✓ interpret images → generate text (captioning)
 - ✓ read text → reason about images (visual QA)
 - ✓ hear audio → animate faces (viseme generation)
 - ✓ combine image + text → improve translation accuracy



Encoder–Decoder Models

Seq2Seq (LSTM-based)

Attention mechanism (Bahdanau / Luong)

Transformers:

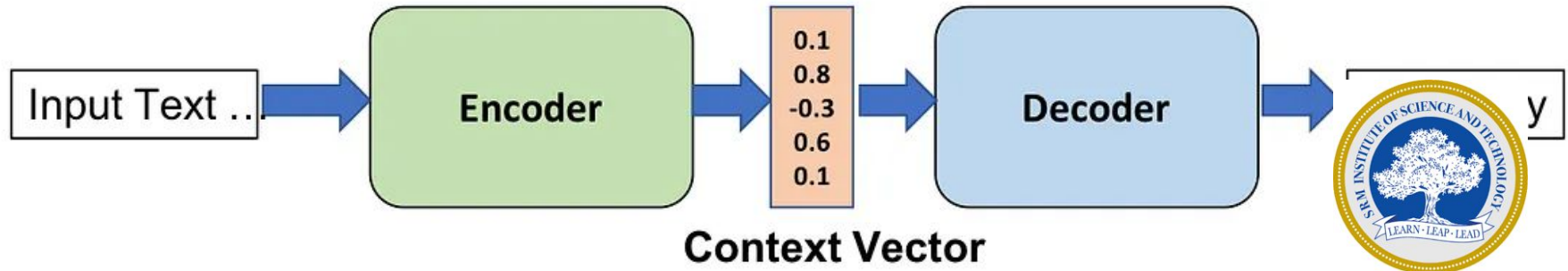
- Encoder
- Decoder
- Cross-attention

Multimodal Transformer architecture

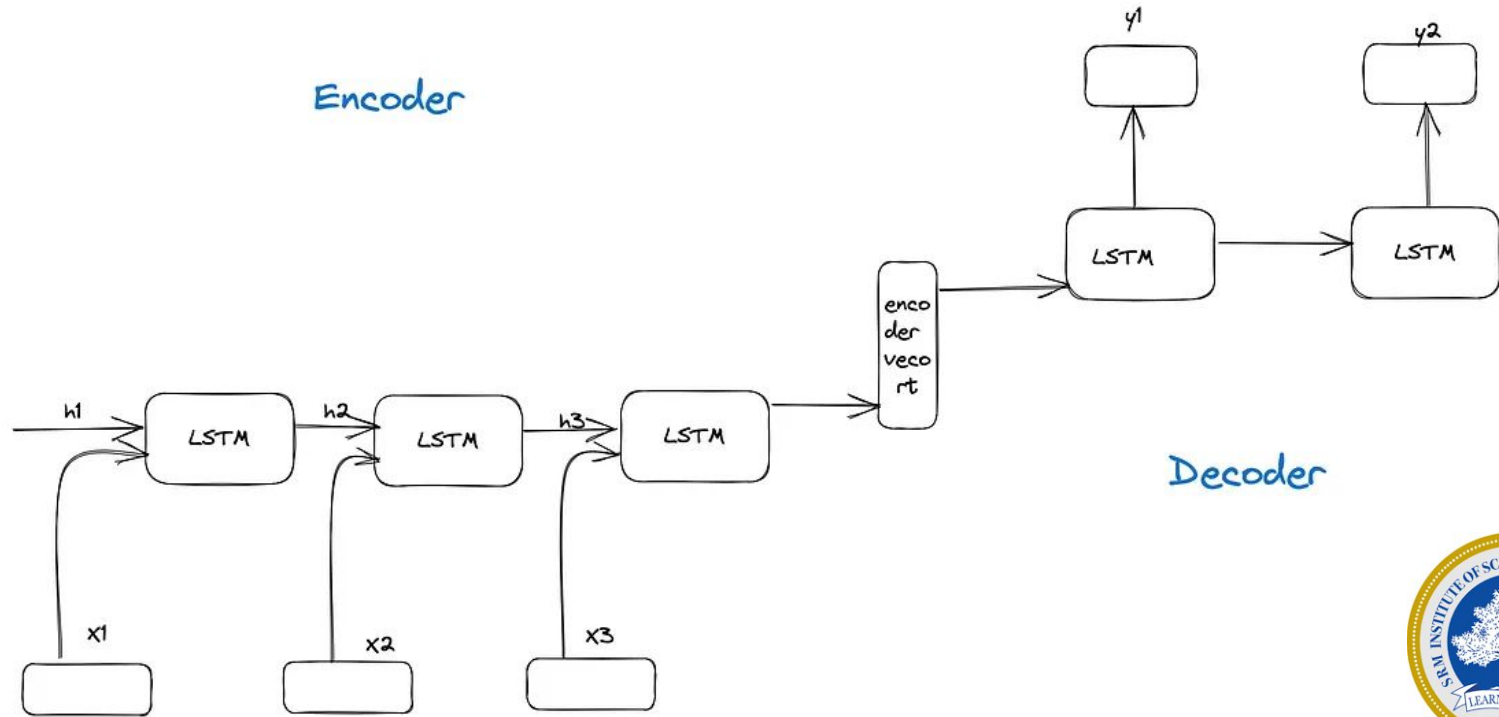


Encoder-Decoder Model

An encoder-decoder is a neural network architecture commonly used in sequence-to-sequence (Seq2Seq) models, particularly in tasks involving natural language processing (NLP) and machine translation. It consists of two main components: an encoder and a decoder.



Seq-Seq





A MICHAEL BAY FILM

TRANSFORMERS

REVENGE OF THE FALLEN

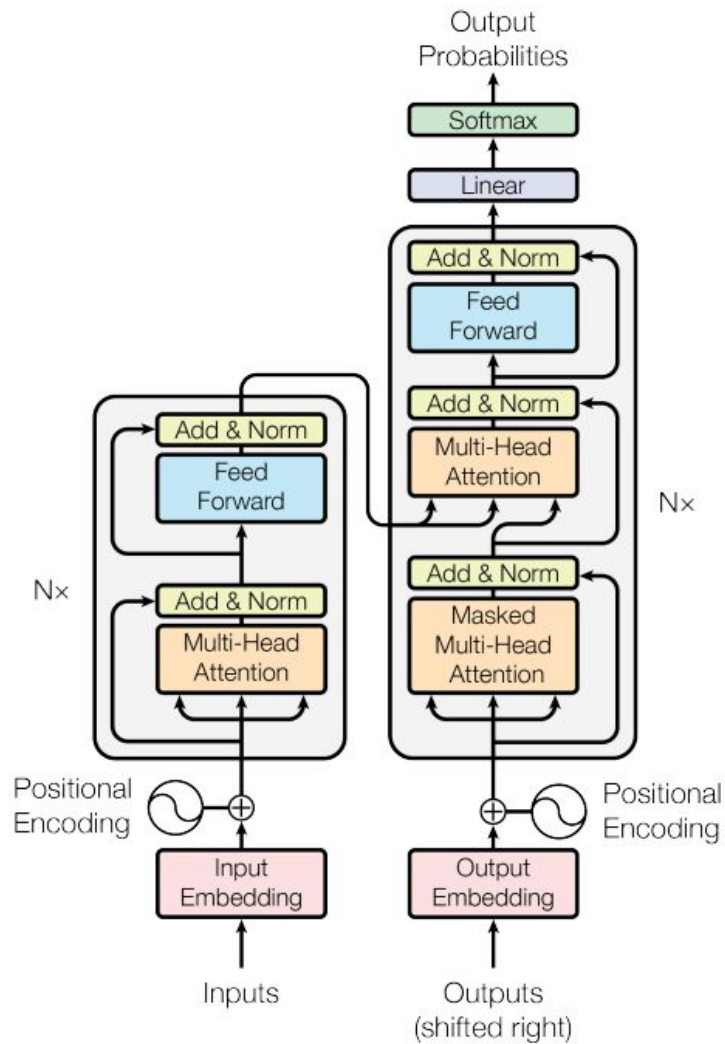
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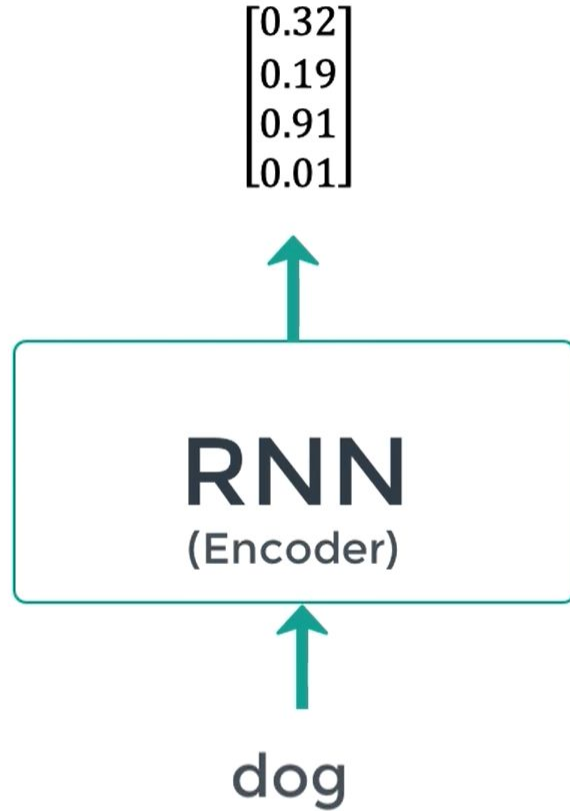
Transformer – Attention is all you Need

- Why use RNN if you need only the attention to take care of the problem.
- This to works on same idea of encoding and decoding
- But this time all the data are encoded at the same time

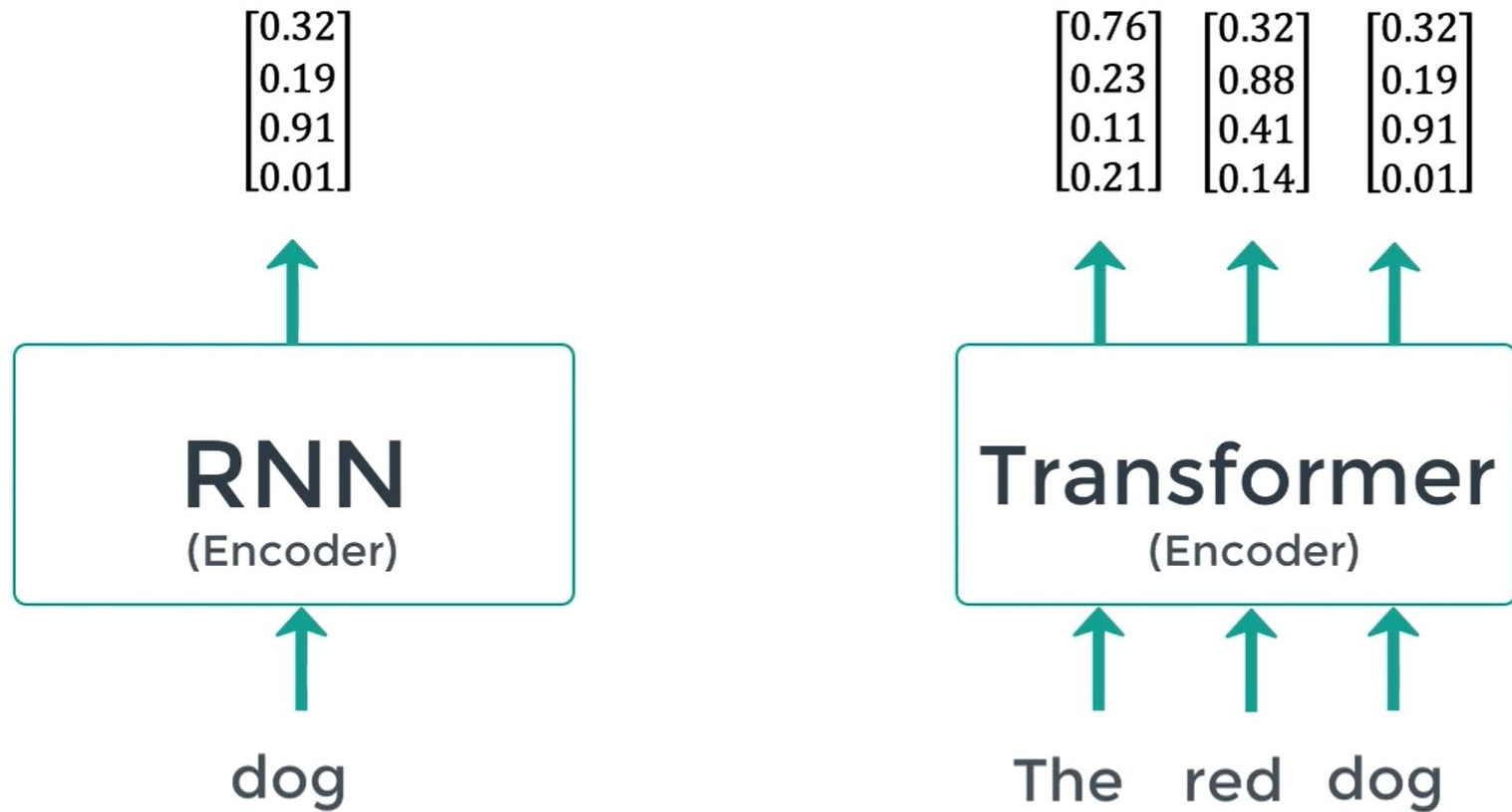




English-French Translation

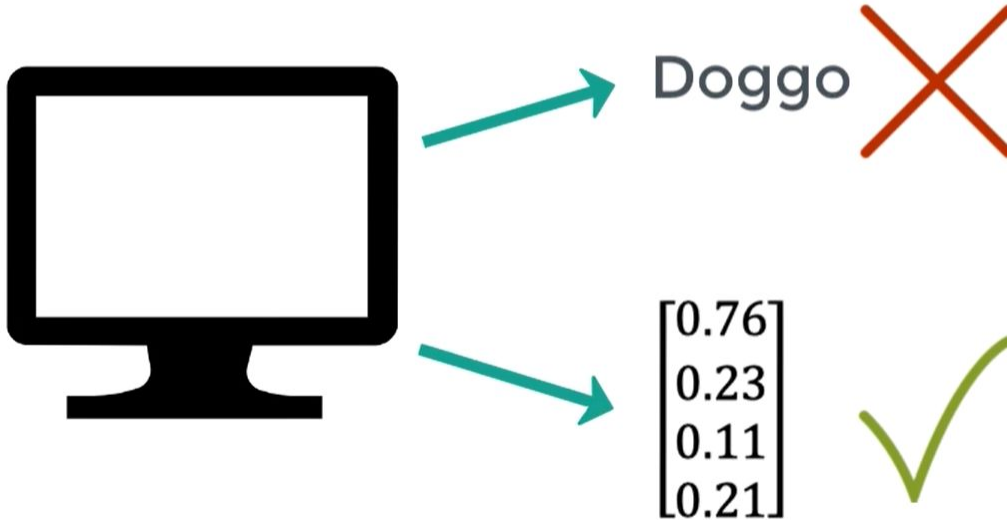


English-French Translation



Transformer Components

Input Embedding



Transformer Components

Input Embedding



learns
→



*“Embedding
Space”*



Transformer Components

Input Embedding



learns
→



*“Embedding
Space”*



Transformer Components

Input Embedding

dog


$$\begin{bmatrix} 0.37 \\ 0.99 \\ 0.01 \\ 0.08 \end{bmatrix}$$

AJ's **dog** is a cutie

AJ looks like a **dog**

Transformer Components

Positional Encoder :vector that gives context based on position of word in sentence

AJ's **dog** is a cutie → Position 2

AJ looks like a **dog** → Position 5

Transformer Components

Positional Encoder : vector that gives context based on position of word in sentence

AJ's **dog** is a cutie → Position 2

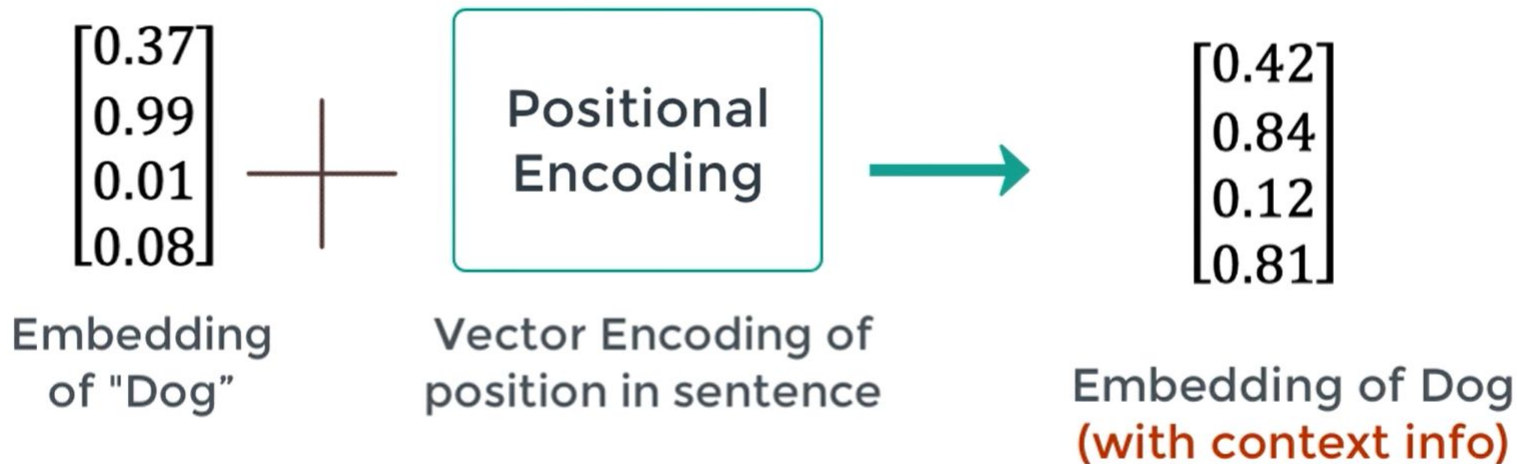
AJ looks like a **dog** → Position 5

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{\text{model}}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}})$$

Transformer Components

Positional Encoder :vector that gives context based on position of word in sentence

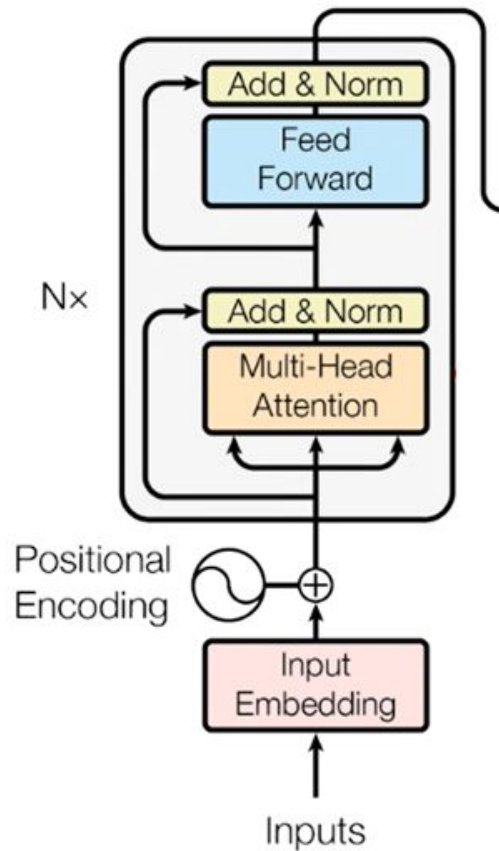


$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{\text{model}}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}})$$

Transformer Components

Encoder Block



Transformer Components

Attention : What part of the input should we focus?

The → The big red dog

big → The big red dog

red → The big red dog

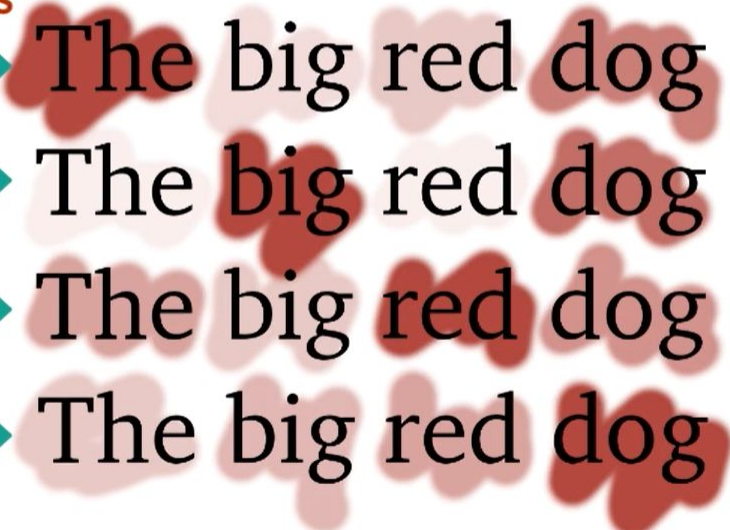
dog → The big red dog

Transformer Components

Attention : What part of the input should we focus?

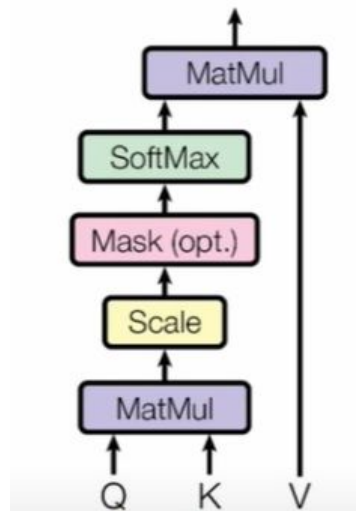
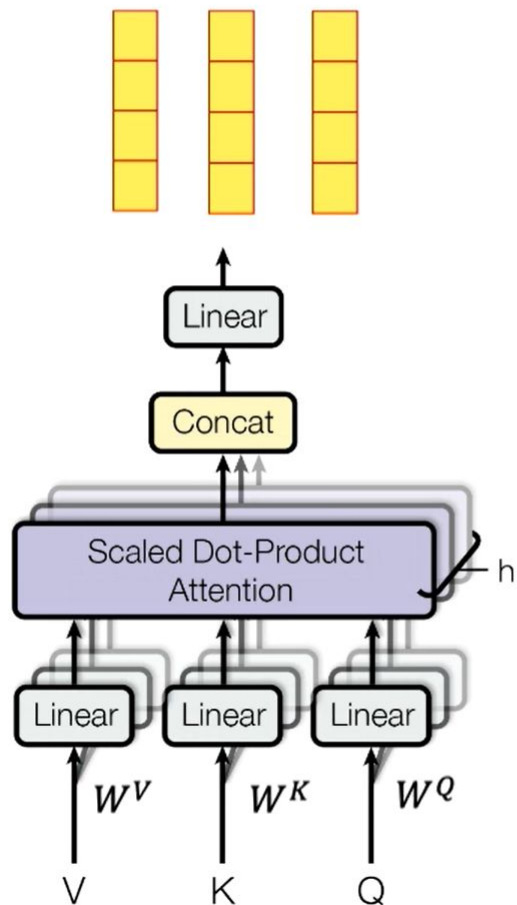
Focus

The → The big red dog
big → The big red dog
red → The big red dog
dog → The big red dog



Attention Vectors

$[0.71 \quad 0.04 \quad 0.07 \quad 0.18]^T$
 $[0.01 \quad 0.84 \quad 0.02 \quad 0.13]^T$
 $[0.09 \quad 0.05 \quad 0.62 \quad 0.24]^T$
 $[0.03 \quad 0.03 \quad 0.03 \quad 0.91]^T$

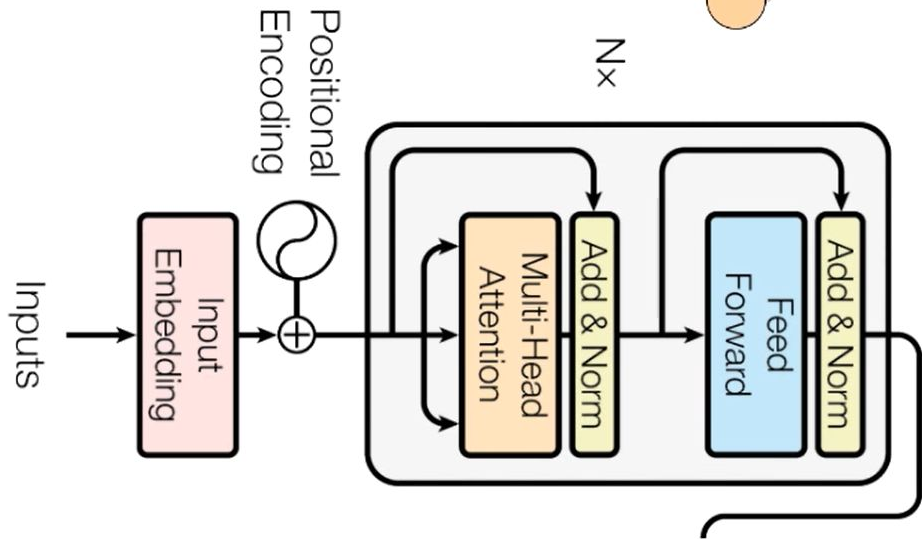
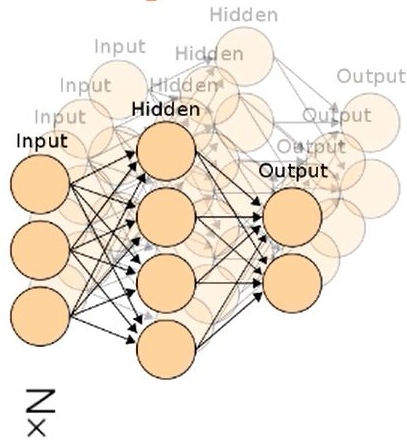


$$Z = \text{softmax} \left(\frac{Q \cdot K^T}{\sqrt{\text{Dimension of vector } Q, K \text{ or } V}}} \right) \cdot V$$

Transformer Components

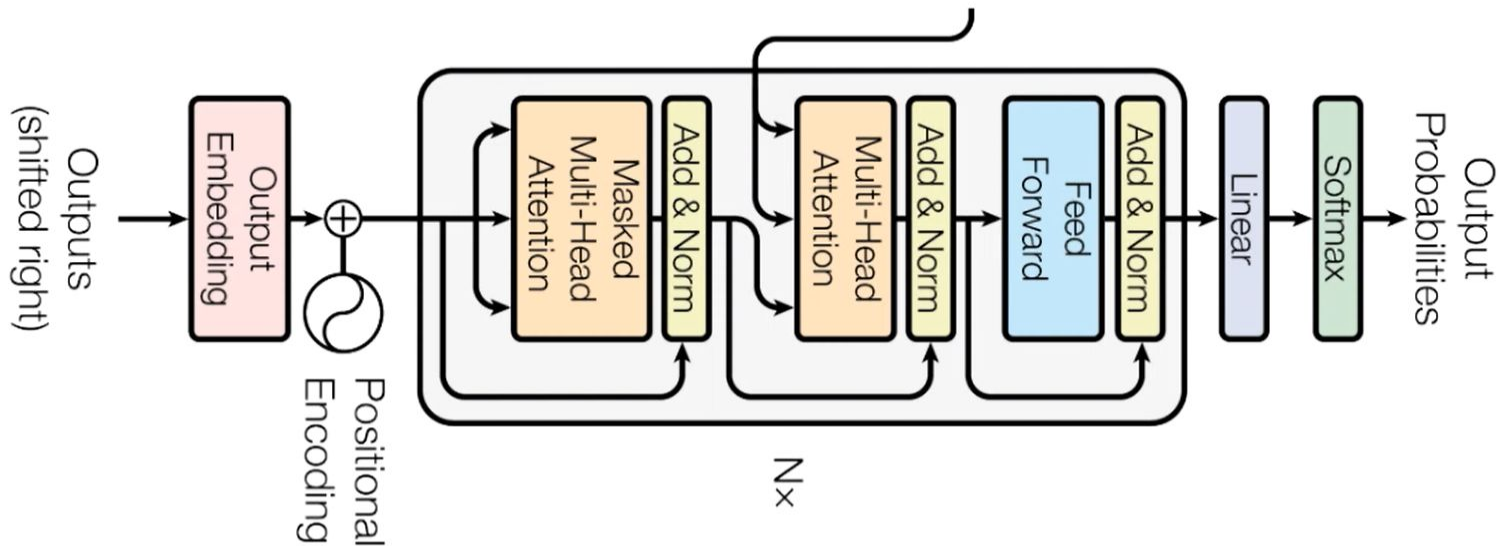
The big red dog

$\begin{bmatrix} 0.71 \\ 0.04 \\ 0.07 \\ 0.18 \end{bmatrix}$



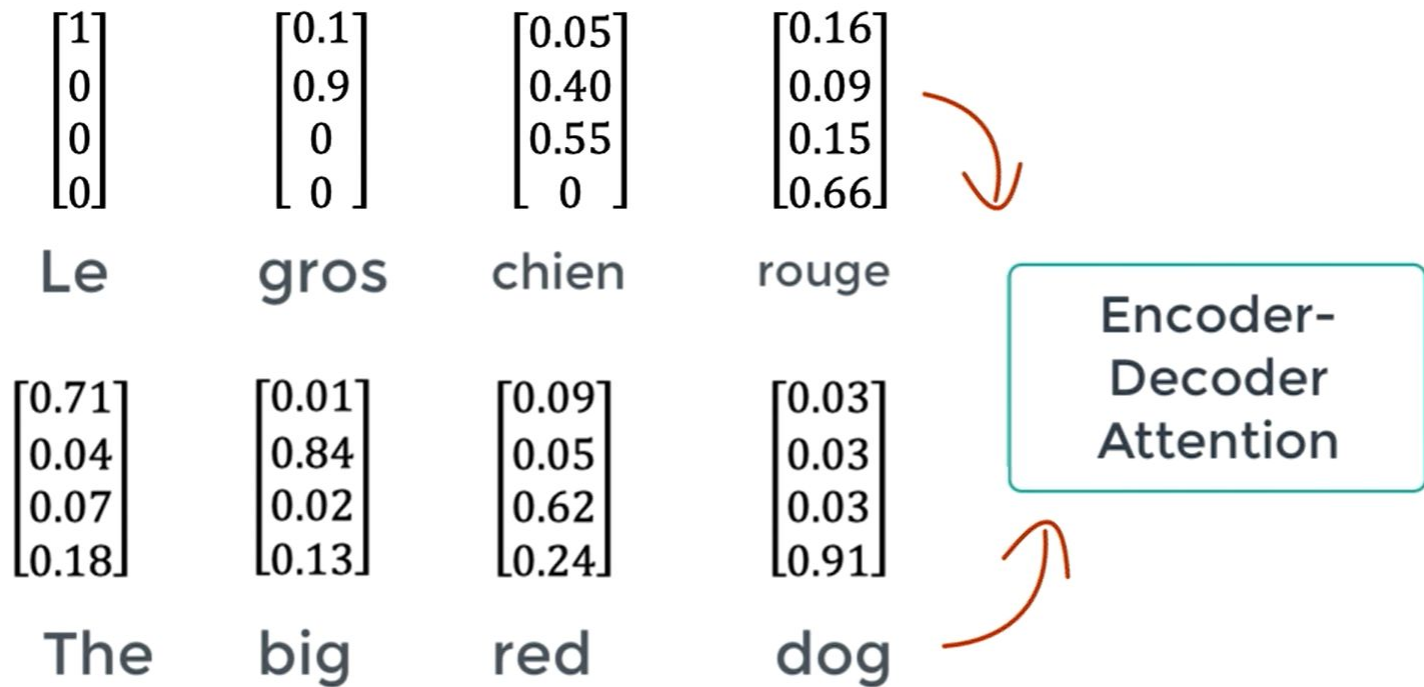
Transformer Components

Decoder



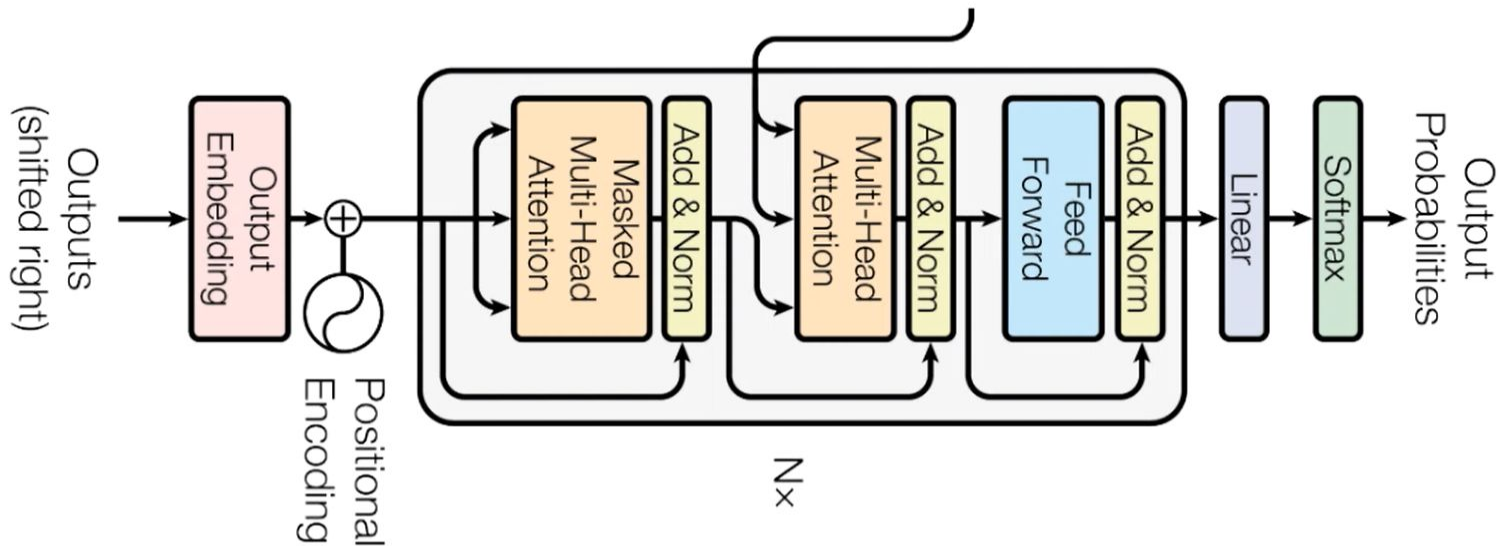
Transformer Components

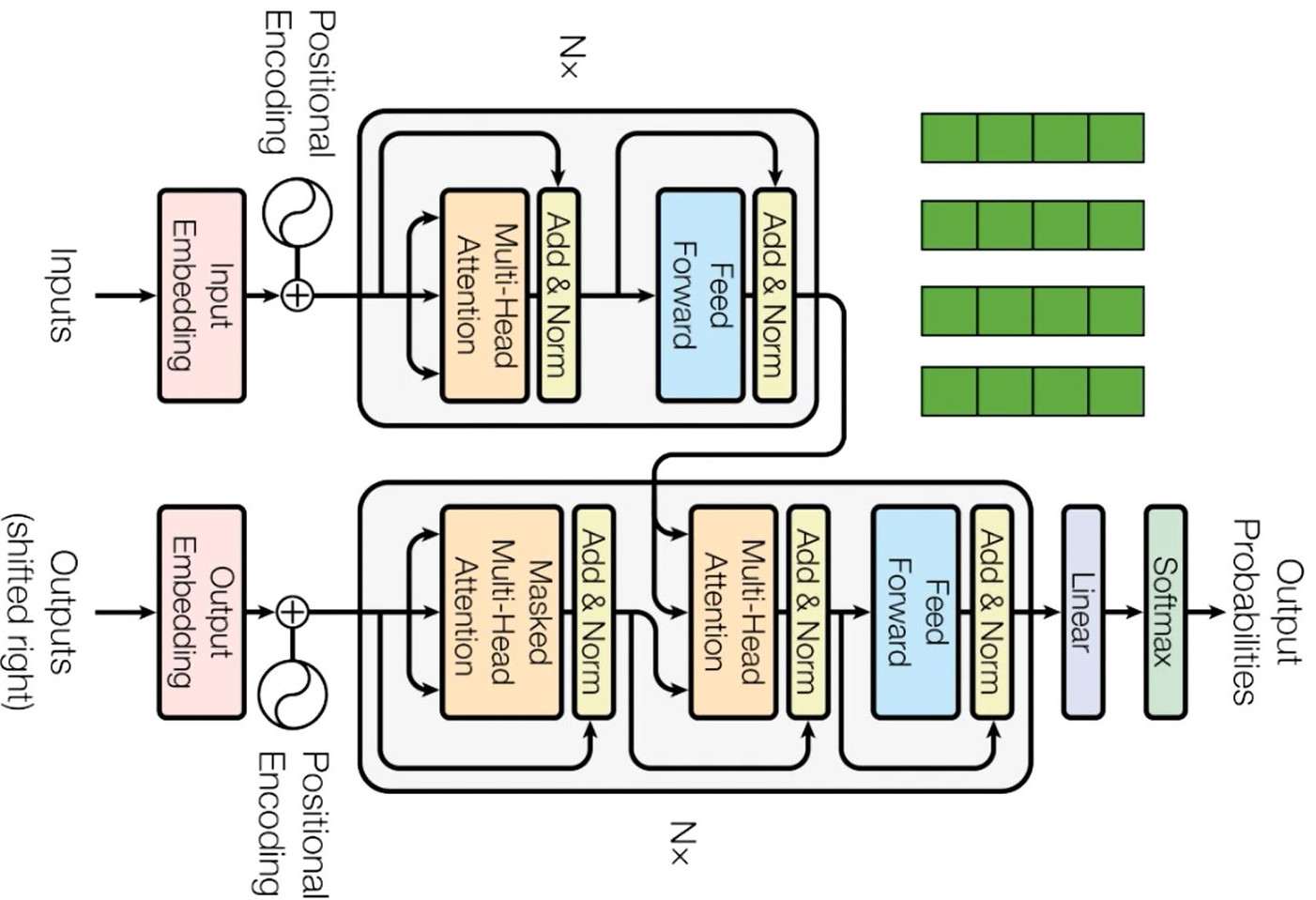
Decoder



Transformer Components

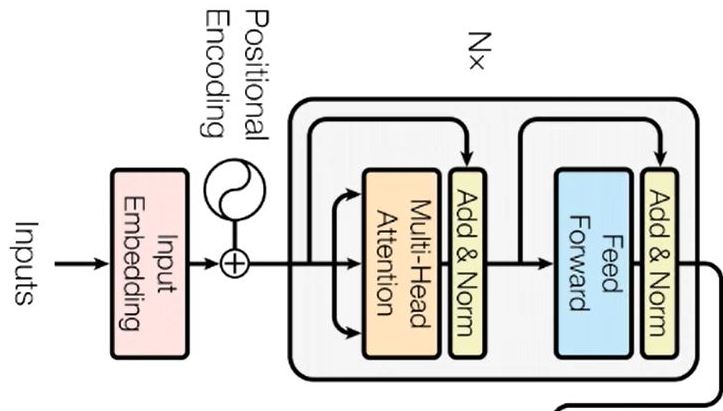
Decoder





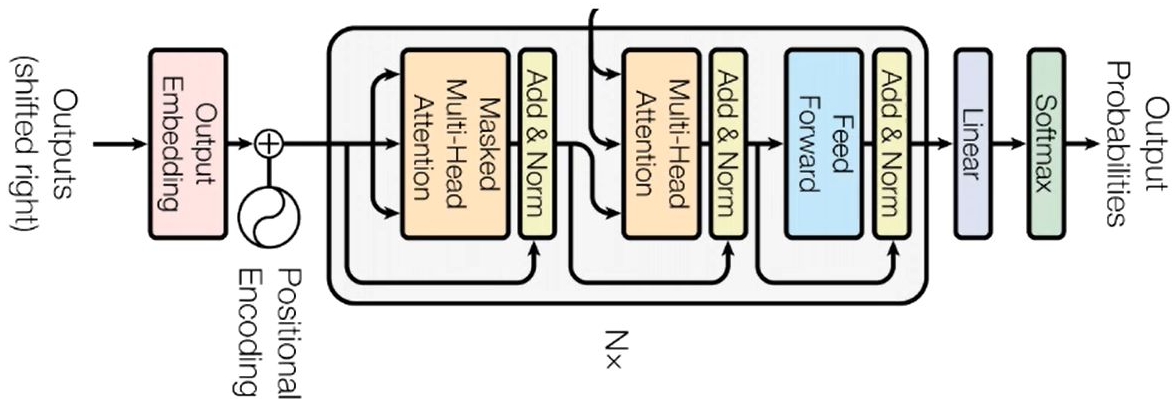
The
big
red
dog

Transformer Flow



What is English? What is context?

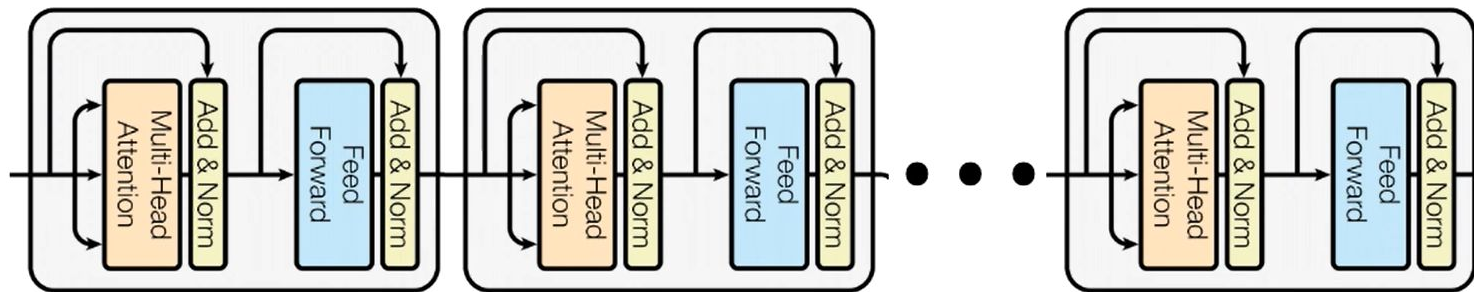
What is language!



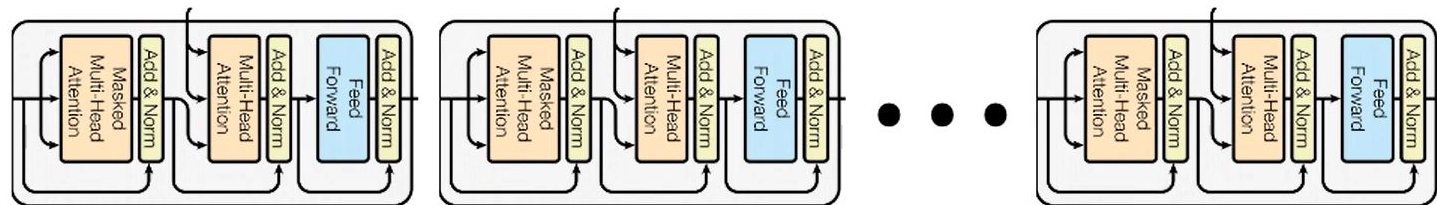
How to map English words to French words?

What is language!

Transformer Flow



BERT



GPT

What Are Multimodal Transformers

Definition:

Multimodal Transformers are models that process **multiple input modalities** (text, image, audio, video) using a unified Transformer architecture.

Goal:

Create a shared representation that allows the model to **reason across modalities**, enabling tasks like captioning, VQA, translation, and generation.

Key Idea:

Different modalities → **encoders** → **aligned latent space** → **cross-attention** → **decoder/output**.

Architecture Overview

1. Per-modality Encoders

- **Text Encoder** → Transformer (e.g., BERT, T5 encoder)
- **Vision Encoder** → ViT / CNN
- **Audio Encoder** → Spectrogram Transformer

2. Fusion Module

- Cross-modal Transformer layers
- Cross-attention between modalities
- Contrastive alignment (CLIP-style)

3. Decoder / Task Head

- Caption generation
- VQA answer decoder
- Translation decoder
- Retrieval scoring head

Fusion Strategies

1. Early Fusion

- Combine modalities at input stage
- Example: Concatenate image patches + text tokens
- Pros: Strong cross-modal interaction
- Cons: Heavy computational cost

2. Late Fusion

- Process separately → fuse at final layers
- Fast and simpler
- But weaker deep interactions

3. Hybrid Fusion (Most Common)

- Separate encoders
- Cross-attention layers integrate modalities
- Used in: Flamingo, LLaVA, Qwen-VL, PaLI-X

Viseme Generation and Visual puppetry

Visemes are the visual shapes of the mouth that appear when a person speaks.

Like phonemes are the smallest sounds in speech, visemes are the lip shapes that represent one or more phonemes.

Example:

The sounds /p/, /b/, and /m/ look almost the same on lips → one viseme.

Dynamic vs. Static: Traditional visemes are static mouth shapes, but newer research focuses on dynamic visemes that capture the fluid motion between sounds to reduce "robotic" transitions.

<https://dan.toronto.edu/~elf/IAI/ISIG16.pdf>

Phoneme set

(1) p, b, m

(2) f, v

(3) t, d, n, l

(4) ay, ey

...

(16) ay, ey

(17) ow, oy

(18) er

Viseme-driven frames from a reference image

Example 1

After

/æ f t er/



Example 2

China

/ch ay n ah/



Example 3

Tomorrow

/t ah m aa r ow/



What is Modality Alignment?

Definition:

Modality alignment is the process of mapping data from **different modalities** (text, image, audio, video) into a **shared, comparable representation**.

Why it matters:

- Enables multimodal tasks: captioning, VQA, speech-driven video, cross-modal retrieval
- Allows models to *understand relationships* across senses—like linking an image to the sentence describing it
- Forms the foundation of models like **CLIP, BLIP, LLaVA, Qwen-VL, Flamingo**

Types of Modality Alignment

1. Latent Alignment (Implicit)

- Align modalities **in embedding space**
- Example: CLIP learns that *“a dog running”* $\leftrightarrow$ *that image*
- No explicit sequence matching

2. Explicit Alignment

- Align sequences **element-by-element**
- Example: DTW aligns **speech phonemes** $\leftrightarrow$ **viseme video frames**

Latent Alignment Approaches

We project each modality into a **shared latent space**, making them comparable.

Image Encoder $\rightarrow$ **z_img**

Text Encoder $\rightarrow$ **z_text**

Train them so:

`cosine_similarity(z_img, z_text)` is **high** for matched pairs.

Examples

- CLIP (image $\leftrightarrow$ text)
- BLIP's pre-training
- Qwen-VL image-text contrastive stage

Strengths

- Scales well to billions of pairs
- Good for retrieval tasks
- No need for temporal alignment

Explicit Alignment

Why explicit?

For time-series / sequence alignment tasks, like:

- **audio ↔ video lip movements**
- speech phonemes ↔ visemes
- gesture ↔ sound alignment
- sign language recognition
- temporal segmentation

Most common explicit method:

★ **Dynamic Time Warping (DTW)**

DTW

Dynamic Time Warping aligns two sequences that may be out of sync in time—e.g., the same gesture performed at different speeds, or speech spoken faster vs slower. Unlike Euclidean distance, which compares point-by-point at the same timestamps, DTW **warps the time axis** so that semantically similar parts line up even if they occur at different rates.

DTW (Dynamic Time Warping)

Align two sequences that may have different lengths or speeds:

Example:

Audio (fast) → 50 frames

Lip video (slow) → 120 frames

DTW finds an optimal **warping path** that minimizes cumulative distance.

Applications

- Speech recognition
- Speaker verification
- Audio-lip synchronization
- Sensor alignment
- Handwriting recognition